Supporting Information

Low Band Gap Fused Bicyclic Polymers with Heteroatoms Se and Te: A DFT-PBC Study

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Table S1. The sum of percent contributions of atomic p orbitals for the frontier orbitals of the monomers and polymers (The sum of atomic p orbitals with threshold of absolute value greater than 0.5 %, the suffix p indicates planar form.)

Polymer	Orbital	p_x %	p_y %	p_z %	Polymer	Orbital	p_x %	p_y %	p_z %
SeSe	HOMO	0.0	0.0	98.3	PSeTe46	HOCO	1.5	11.4	80.8
SeSe	LUMO	0.0	0.0	97.4	PSeTe46	LUCO	0.0	14.4	76.4
SeTe	HOMO	0.0	0.0	99.0	PSeTe46p	HOCO	0.0	0.0	97.3
SeTe	LUMO	51.3	38.5	0.0	PSeTe46p	LUCO	0.0	0.0	96.6
TeSe	HOMO	0.0	0.0	98.2	PTeSe24	HOCO	0.0	0.0	95.1
TeSe	LUMO	0.0	0.0	97.7	PTeSe24	LUCO	0.0	0.0	95.0
TeTe	HOMO	0.0	0.0	98.9	PTeSe24p	HOCO	0.0	0.0	97.1
TeTe	LUMO	43.7	46.9	0.0	PTeSe24p	LUCO	0.0	0.0	98.0
PSeSe24	HOCO	0.0	0.0	96.4	PTeSe26	HOCO	0.0	3.4	89.2
PSeSe24	LUCO	0.0	0.0	96.7	PTeSe26	LUCO	0.0	6.8	86.8
PSeSe24p	HOCO	0.0	0.0	97.1	PTeSe26p	HOCO	0.0	0.0	98.1
PSeSe24p	LUCO	0.0	0.0	97.8	PTeSe26p	LUCO	0.0	0.0	98.0
PSeSe26	HOCO	0.0	0.5	93.1	PTeSe36	HOCO	0.0	26.3	67.1
PSeSe26	LUCO	0.0	1.2	93.6	PTeSe36	LUCO	4.2	27.0	56.8
PSeSe26p	HOCO	0.0	0.0	98.1	PTeSe36p	HOCO	0.0	0.0	98.6
PSeSe26p	LUCO	0.0	0.0	98.0	PTeSe36p	LUCO	0.0	0.0	97.3
PSeSe36	HOCO	0.0	23.3	69.2	PTeSe46	HOCO	4.8	11.9	75.3
PSeSe36	LUCO	0.0	21.0	68.4	PTeSe46	LUCO	0.0	19.9	73.4
PSeSe36p	HOCO	0.0	0.0	98.5	PTeSe46p	HOCO	0.0	0.0	97.7
PSeSe36p	LUCO	0.0	0.0	97.3	PTeSe46p	LUCO	0.0	0.0	97.4
PSeSe46	HOCO	0.0	7.7	86.4	PTeTe24	HOCO	0.0	0.0	93.2
PSeSe46	LUCO	0.0	8.6	84.2	PTeTe24	LUCO	0.0	2.0	91.4
PSeSe46p	HOCO	0.0	0.0	96.9	PTeTe24p	HOCO	0.0	0.0	97.5
PSeSe46p	LUCO	0.0	0.0	95.6	PTeTe24p	LUCO	0.0	0.0	97.9
PSeTe24	HOCO	0.0	0.0	94.0	PTeTe26	HOCO	0.0	7.7	84.1
PSeTe24	LUCO	0.0	1.4	92.3	PTeTe26	LUCO	26.2	37.0	28.3
PSeTe24p	HOCO	0.0	0.0	97.5	PTeTe26p	HOCO	0.0	0.0	98.1
PSeTe24p	LUCO	0.0	0.0	97.7	PTeTe26p	LUCO	51.0	36.0	0.0
PSeTe26	HOCO	0.0	3.3	87.9	PTeTe36	HOCO	0.5	28.0	63.6
PSeTe26	LUCO	0.0	3.5	90.1	PTeTe36	LUCO	14.1	37.8	38.5
PSeTe26p	HOCO	0.0	0.0	97.3	PTeTe36p	HOCO	0.0	0.0	99.0
PSeTe26p	LUCO	0.0	0.0	98.0	PTeTe36p	LUCO	0.0	0.0	97.6
PSeTe36	HOCO	0.6	24.0	66.7	PTeTe46	HOCO	5.6	19.2	68.5
PSeTe36	LUCO	2.5	29.9	57.8	PTeTe46	LUCO	0.0	27.5	65.2
PSeTe36p	HOCO	0.0	0.0	98.9	PTeTe46p	HOCO	0.0	0.0	95.7
PSeTe36p	LUCO	0.0	0.0	97.6	PTeTe46p	LUCO	0.0	0.0	97.5

Table S2. Some properties of frontier orbitals of the polymers (X, Y are heteroatoms of left monomer and X', Y' are heteroatoms of right monomer in the cell given in Fig. 5, the suffix p indicates planar form.)

Polymer	Orbital	E (eV)	E_g (eV)	s %	p %	d %	ODI	X %	Y %	X' %	Y' %
PSeSe24	HOCO	-4.87		0.01	98.22	1.76	11.04	0.44	3.53	0.44	3.53
PSeSe24	LUCO	-3.22	1.65	0.03	98.23	1.74	7.71	8.48	2.39	8.48	2.39
PSeSe24p	HOCO	-4.85		0.00	98.24	1.76	11.10	0.40	3.52	0.40	3.52
PSeSe24p	LUCO	-3.23	1.63	0.00	98.26	1.74	7.74	8.46	2.28	8.45	2.28
PSeSe26	HOCO	-5.39		0.10	98.65	1.24	10.14	3.79	5.64	3.80	5.63
PSeSe26	LUCO	-2.35	3.04	0.01	98.46	1.53	11.32	8.73	0.07	8.73	0.07
PSeSe26p	HOCO	-5.40		0.00	98.76	1.24	10.53	3.69	4.94	3.69	4.94
PSeSe26p	LUCO	-2.39	3.01	0.00	98.49	1.51	11.50	8.65	0.08	8.65	0.08
PSeSe36	HOCO	-5.31		0.33	98.52	1.14	10.40	1.31	16.97	1.32	16.95
PSeSe36	LUCO	-1.81	3.50	0.50	98.15	1.35	9.69	6.95	1.08	6.95	1.07
PSeSe36p	HOCO	-5.32		0.00	98.95	1.06	11.27	0.12	17.99	0.12	17.98
PSeSe36p	LUCO	-2.28	3.04	0.00	98.75	1.25	10.21	6.04	0.54	6.04	0.54
PSeSe46	HOCO	-4.49		0.00	98.34	1.66	8.79	0.63	8.54	0.63	8.54
PSeSe46	LUCO	-3.27	1.22	0.58	97.58	1.84	8.86	9.98	0.60	9.98	0.60
PSeSe46p	HOCO	-4.26		0.00	98.23	1.77	9.47	0.33	7.38	0.33	7.38
PSeSe46p	LUCO	-3.48	0.78	0.00	98.29	1.71	10.61	10.89	1.09	10.88	1.10
PSeTe24	HOCO	-4.83		0.12	98.22	1.66	10.42	0.47	5.06	0.46	5.06
PSeTe24	LUCO	-3.21	1.63	0.13	98.18	1.69	7.56	8.09	3.16	8.09	3.17
PSeTe24p	HOCO	-4.77		0.00	98.33	1.67	10.52	0.27	5.21	0.27	5.21
PSeTe24p	LUCO	-3.22	1.54	0.00	98.28	1.72	7.76	8.09	2.12	8.08	2.12
PSeTe26	HOCO	-5.31		0.17	98.68	1.14	9.65	3.36	9.12	3.36	9.12
PSeTe26	LUCO	-2.23	3.08	0.21	98.28	1.51	11.07	8.38	0.39	8.38	0.39
PSeTe26p	HOCO	-5.35		0.00	98.84	1.17	10.27	3.21	6.92	3.22	6.92
PSeTe26p	LUCO	-2.31	3.03	0.00	98.53	1.47	11.48	8.54	0.14	8.54	0.14
PSeTe36	HOCO	-5.19		0.35	98.66	0.98	12.52	1.23	21.26	1.23	21.26
PSeTe36	LUCO	-1.66	3.53	1.07	97.67	1.25	8.85	7.40	3.53	7.39	3.51
PSeTe36p	HOCO	-5.24		0.00	99.13	0.87	13.60	0.04	22.45	0.04	22.47
PSeTe36p	LUCO	-2.19	3.05	0.00	98.77	1.23	10.21	5.95	0.41	5.95	0.41
PSeTe46	HOCO	-4.63		0.01	98.51	1.48	8.89	1.03	10.95	1.03	10.95
PSeTe46	LUCO	-3.12	1.51	0.78	97.35	1.87	8.36	8.64	1.54	8.64	1.54
PSeTe46p	HOCO	-4.30		0.00	98.33	1.67	9.30	0.21	8.81	0.21	8.81
PSeTe46p	LUCO	-3.39	0.91	0.00	98.31	1.69	10.22	10.60	1.32	10.60	1.32

Table S2. (cont'd) Some properties of frontier orbitals of the polymers (X, Y are heteroatoms of left monomer and X', Y' are heteroatoms of right monomer in the cell given in Fig. 5, the suffix p indicates planar form.)

Polymer	Orbital	E (eV)	E_g (eV)	s %	p %	d %	ODI	X %	Y %	X' %	Y' %
PTeSe24	HOCO	-4.76		0.06	98.20	1.74	11.04	0.56	3.68	0.57	3.67
PTeSe24	LUCO	-3.25	1.51	0.06	98.28	1.66	7.78	9.12	2.36	9.12	2.35
PTeSe24p	HOCO	-4.72		0.00	98.26	1.74	11.16	0.44	3.70	0.44	3.72
PTeSe24p	LUCO	-3.27	1.46	0.00	98.33	1.67	7.82	8.99	2.10	8.99	2.10
PTeSe26	HOCO	-5.26		0.16	98.64	1.19	9.37	5.65	5.67	5.66	5.66
PTeSe26	LUCO	-2.34	2.92	-0.03	98.58	1.46	11.10	9.65	0.20	9.65	0.20
PTeSe26p	HOCO	-5.27		0.00	98.80	1.20	9.98	5.25	4.59	5.25	4.59
PTeSe26p	LUCO	-2.39	2.88	0.00	98.54	1.46	11.53	8.65	0.13	8.65	0.13
PTeSe36	HOCO	-5.19		0.39	98.50	1.11	9.86	2.57	16.43	2.57	16.43
PTeSe36	LUCO	-1.79	3.40	0.93	97.88	1.19	9.25	10.37	1.55	10.37	1.55
PTeSe36p	HOCO	-5.26		0.00	98.97	1.04	11.49	0.10	18.60	0.10	18.59
PTeSe36p	LUCO	-2.34	2.92	0.00	98.78	1.22	10.08	5.69	0.55	5.69	0.55
PTeSe46	HOCO	-4.72		0.06	98.42	1.51	8.69	2.44	8.63	2.44	8.64
PTeSe46	LUCO	-2.93	1.78	0.43	97.68	1.89	8.94	10.04	0.39	10.05	0.39
PTeSe46p	HOCO	-4.25		0.00	98.40	1.60	10.39	12.42	0.21	12.42	0.21
PTeSe46p	LUCO	-3.50	0.75	0.00	98.27	1.73	11.03	2.18	7.10	2.18	7.10
PTeTe24	HOCO	-4.74		0.20	98.15	1.65	10.47	0.62	4.96	0.60	4.96
PTeTe24	LUCO	-3.28	1.46	0.11	98.27	1.62	7.62	8.78	3.08	8.77	3.13
PTeTe24p	HOCO	-4.67		0.00	98.35	1.65	10.44	0.28	5.46	0.28	5.46
PTeTe24p	LUCO	-3.29	1.37	0.00	98.34	1.66	7.82	8.65	1.86	8.65	1.86
PTeTe26	HOCO	-5.19		0.23	98.67	1.10	8.91	5.19	9.29	5.19	9.29
PTeTe26	LUCO	-2.26	2.94	2.93	96.22	0.84	10.20	17.32	9.18	17.32	9.19
PTeTe26p	HOCO	-5.24		0.00	98.87	1.13	9.75	4.63	6.38	4.63	6.38
PTeTe26p	LUCO	-2.53	2.72	8.48	91.16	0.36	13.87	17.96	18.19	17.96	18.19
PTeTe36	HOCO	-5.07		0.37	98.67	0.96	11.95	2.26	20.69	2.25	20.71
PTeTe36	LUCO	-1.71	3.36	2.46	96.59	0.94	8.79	13.59	5.94	13.76	5.98
PTeTe36p	HOCO	-5.17		0.00	99.15	0.85	14.03	0.02	23.19	0.02	23.18
PTeTe36p	LUCO	-2.25	2.93	0.00	98.81	1.19	10.06	5.62	0.40	5.63	0.40
PTeTe46	HOCO	-4.84		0.08	98.59	1.33	8.91	2.82	11.81	2.82	11.82
PTeTe46	LUCO	-2.77	2.07	0.42	97.67	1.90	8.70	9.21	1.09	9.21	1.09
PTeTe46p	HOCO	-4.26		0.00	98.30	1.70	10.43	0.71	7.29	0.70	7.29
PTeTe46p	LUCO	-3.53	0.73	0.00	98.47	1.53	11.78	11.15	2.33	11.15	2.33

Table S3. The band gap, HOCO and LUCO values (in eV) calculated by B3PW91/SDD method.

Band gap (eV)			
Y/X	Se	Te	Po
Se	0.766	0.723	0.813
Te	0.885	0.717	0.663
Po	0.919	0.764	0.686
HOCO (eV)			
Y/X	Se	Te	Po
Se	-4.358	-4.366	-4.330
Te	-4.433	-4.409	-4.310
Po	-4.333	-4.331	-4.246
LUCO (eV)			
Y/X	Se	Te	Po
Se	-3.592	-3.644	-3.517
Te	-3.548	-3.693	-3.647
Po	-3.413	-3.567	-3.561

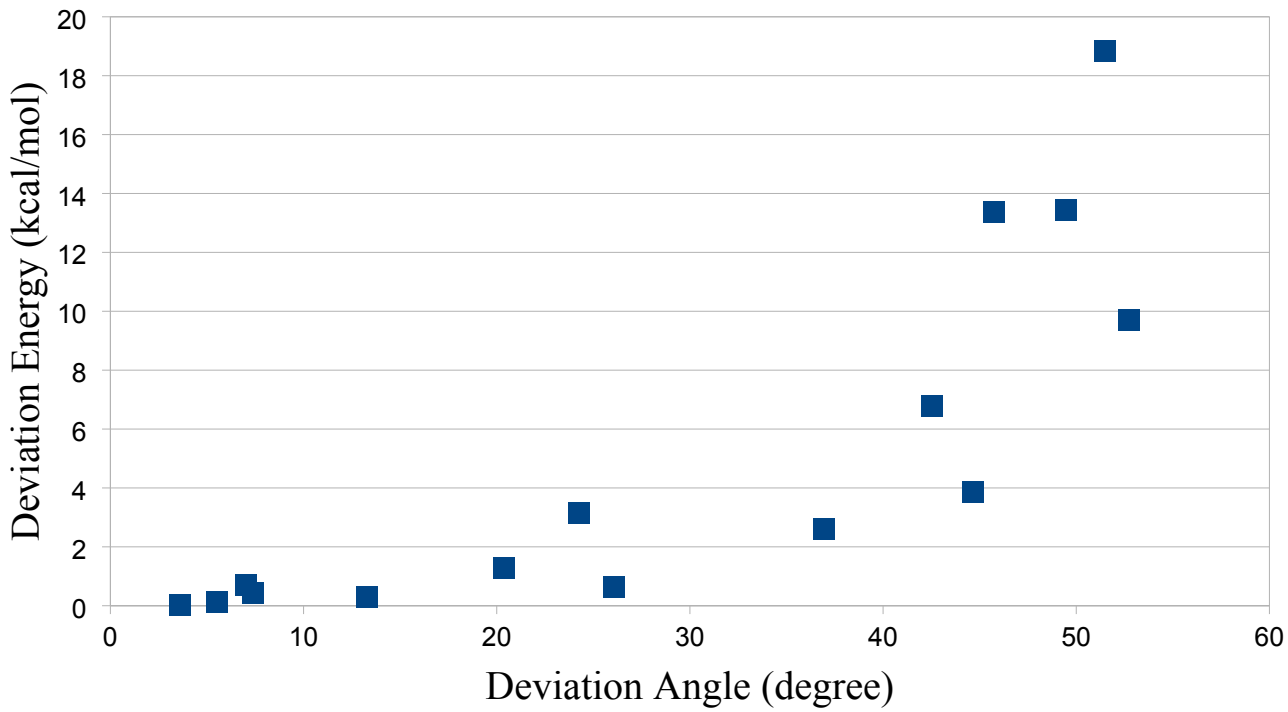
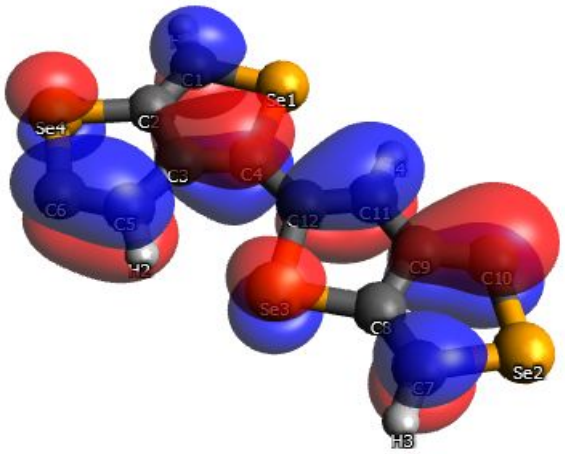
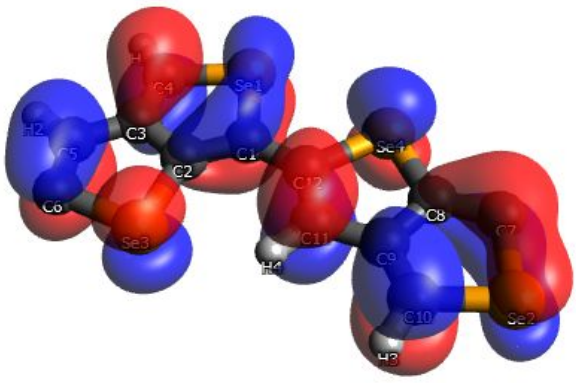


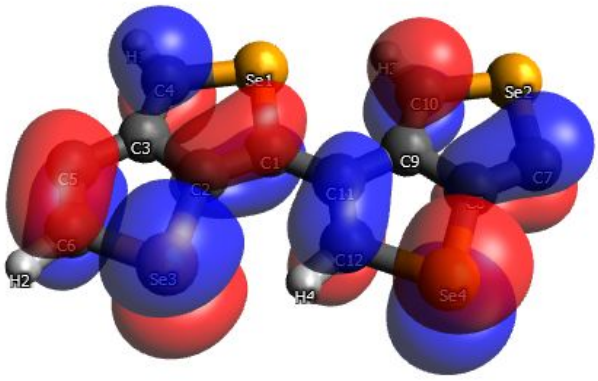
Figure S1. The deviation energies (E_{planar}-E_{nr}) versus the deviation angle (from 180^{o}).



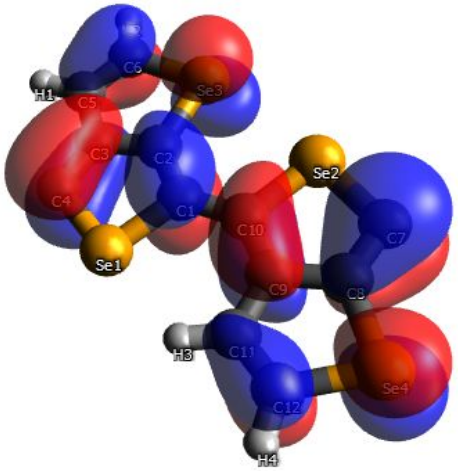
PSeSe24



PSeSe26

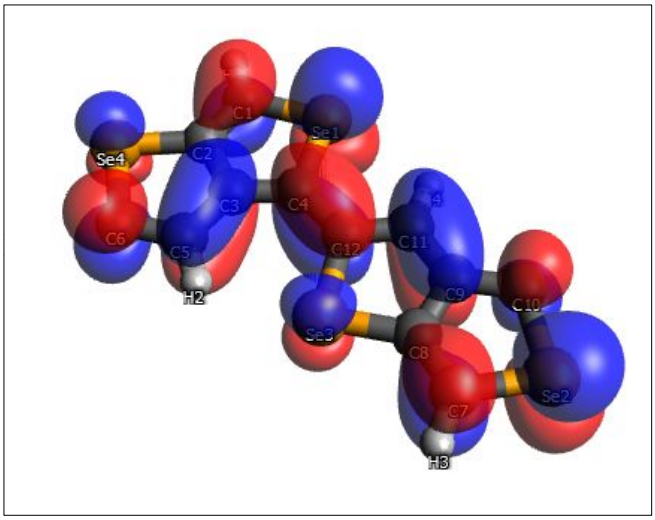


PSeSe36

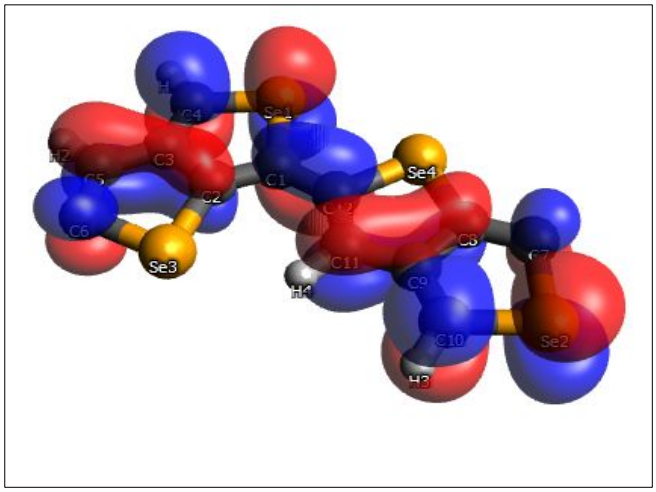


PSeSe46

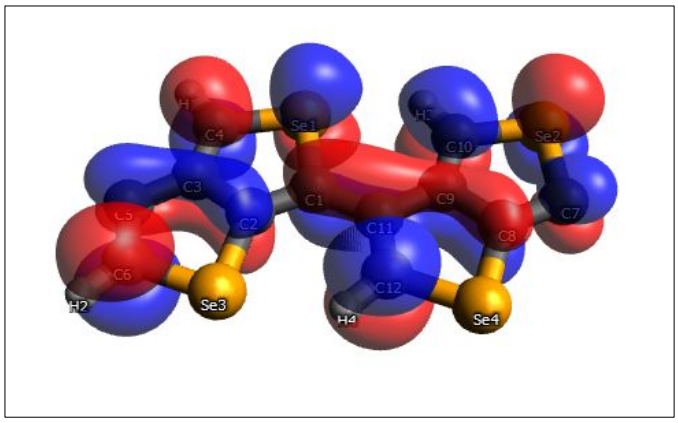
Figure S2. HOCOs of PSeSe for planar geometry.



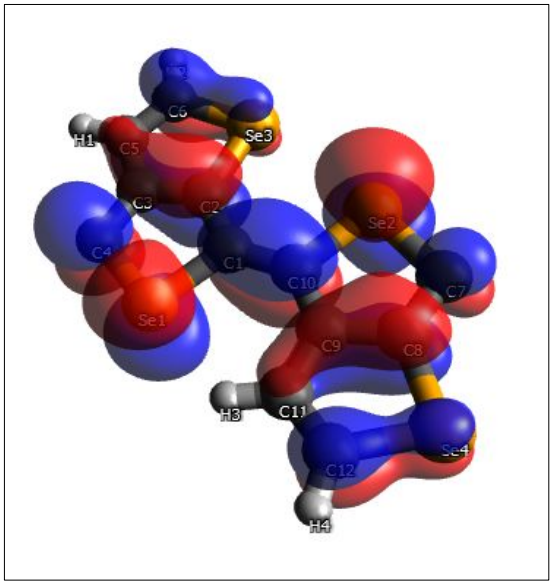
PSeSe24



PSeSe26



PSeSe36



PSeSe46

Figure S3. LUCOs of PSeSe for planar geometry.